

Lecture 7 Tutorial — Support Vector Machine

Q1. Signed Distance to a Hyperplane (Derivation)

Let the hyperplane be

$$H = \{x : \mathbf{w}^\top x + b = 0\}, \quad \mathbf{w} \neq 0.$$

1. Write the signed distance from a point x to H .
2. State what the sign indicates.

Q2. Scale Invariance and the Need for Normalization (Short Proof)

Consider the classifier

$$\hat{y} = \text{sign}(\mathbf{w}^\top x + b).$$

1. Show that scaling $(\mathbf{w}, b) \mapsto (c\mathbf{w}, cb)$ for any $c > 0$ does not change the decision boundary.
2. Show how the quantity $y_i(\mathbf{w}^\top x_i + b)$ scales under $(\mathbf{w}, b) \mapsto (c\mathbf{w}, cb)$.
3. Explain (one sentence) why a margin objective without normalization is not meaningful.

Q3. Geometric Margin (Definition + Reasoning)

For a labeled sample (x_i, y_i) with $y_i \in \{+1, -1\}$, define the geometric margin

$$\gamma_i = \frac{y_i(\mathbf{w}^\top x_i + b)}{\|\mathbf{w}\|}.$$

1. Define the margin of the classifier on the dataset in terms of γ_i .
2. If one defines an “unnormalized margin” as $y_i(\mathbf{w}^\top x_i + b)$ (no $\|\mathbf{w}\|$), what degeneracy occurs under scaling? State the issue precisely.

Q4. Hard-Margin SVM (Primal Formulation)

Write the hard-margin SVM primal optimization problem in the standard normalized form.

- Decision variables: _____
- Objective: _____
- Constraints: _____

Use the normalization

$$y_i(\mathbf{w}^\top x_i + b) \geq 1 \quad \text{for all } i.$$

Q5. Margin Width (Calculation)

Under the hard-margin normalization

$$y_i(\mathbf{w}^\top x_i + b) \geq 1,$$

the two supporting hyperplanes are

$$\mathbf{w}^\top x + b = 1 \quad \text{and} \quad \mathbf{w}^\top x + b = -1.$$

1. Compute the distance between these two hyperplanes in terms of $\|\mathbf{w}\|$.
2. Conclude why minimizing $\|\mathbf{w}\|$ corresponds to maximizing separation.

Q6. Convexity Check (True / False with One-Line Justification)

Write T or F, and justify in one line.

1. The feasible region of the hard-margin SVM primal is convex because each constraint is a linear inequality in $(\mathbf{w}, b)$.
2. The objective $\frac{1}{2}\|\mathbf{w}\|^2$ is convex.
3. If both (1) and (2) are true, any local minimum is also a global minimum.

Q7. Lagrangian and Stationarity (Derivation)

Rewrite the constraints as

$$1 - y_i(\mathbf{w}^\top x_i + b) \leq 0$$

and introduce multipliers

$$\alpha_i \geq 0.$$

1. Write the Lagrangian $L(\mathbf{w}, b, \alpha)$.
2. Compute the stationarity conditions:

$$\frac{\partial L}{\partial \mathbf{w}} = 0, \quad \frac{\partial L}{\partial b} = 0.$$

3. State the resulting expressions for $\mathbf{w}$ and the equality constraint involving α_i and y_i .

Q8. Dual Problem (Write Down + Interpretation)

Using the stationarity results, write the hard-margin SVM dual problem.

- Maximize: _____
- Subject to: _____

Then answer:

1. Which training points can influence $\mathbf{w}$ in the dual representation

$$\mathbf{w} = \sum_i \alpha_i y_i x_i ?$$

2. What does it imply when “only a few α_i are nonzero”?

Q9. KKT Complementary Slackness (Classification of Points)

For hard-margin SVM, complementary slackness gives

$$\alpha_i(1 - y_i(\mathbf{w}^\top x_i + b)) = 0.$$

For each case, state whether $\alpha_i = 0$ or $\alpha_i > 0$:

1. $y_i(\mathbf{w}^\top x_i + b) > 1$
2. $y_i(\mathbf{w}^\top x_i + b) = 1$

Then state the definition: support vectors are exactly the points such that _____.

Q10. Recovering from Support Vectors (Computation)

Assume you have solved the dual and obtained

$$\mathbf{w} = \sum_i \alpha_i y_i x_i.$$

Let k be a support vector index (i.e., $\alpha_k > 0$). Use the margin equality to compute b :

$$b = y_k - \mathbf{w}^\top x_k.$$

Compute b for:

$$\mathbf{w} = (2, -1)^\top, \quad x_k = (1, 2)^\top, \quad y_k = +1.$$

Q11. Soft-Margin SVM and Slack Interpretation (Computation)

Soft-margin constraints:

$$y_i(\mathbf{w}^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

Given three points with the model outputs:

$$y_1 f(x_1) = 1.2, \quad y_2 f(x_2) = 0.7, \quad y_3 f(x_3) = -0.3.$$

1. Compute ξ_1, ξ_2, ξ_3 .
2. Classify each point as:
 - (A) outside/on margin (no violation),
 - (B) inside margin but correctly classified,
 - (C) misclassified.

Q12. Kernel SVM Decision Function (Computation)

Kernel SVM decision function:

$$\hat{y} = \text{sign}(f(x)), \quad f(x) = \sum_{i \in S} \alpha_i y_i K(x_i, x) + b.$$

Assume there are two support vectors with:

$$\begin{aligned} x_1 &= (1, 0), & y_1 &= +1, & \alpha_1 &= 2 \\ x_2 &= (0, 1), & y_2 &= -1, & \alpha_2 &= 1 \\ b &= -0.5 \end{aligned}$$

Use the linear kernel

$$K(x, z) = x^\top z.$$

For

$$x = (2, 1),$$

compute:

1. $f(x)$
2. The predicted label $\hat{y} \in \{+1, -1\}$